WATER RESOURCES, WELLS & PUMPS · NOTE WR06

Tank and Pond Irrigation and Lift Irrigation in India

Course topics: IDE8.16 L6, IDE8.16 L7

Tanks and ponds

- **Tanks** are surface reservoirs that store rainfall runoff from their catchments and collect seepage from the surrounding water-bearing formations. In India they are generally small reservoirs behind earthen dams, used for irrigation, water supply and groundwater recharge. Tanks built only for recharge are **percolation tanks**.
- **Ponds** are smaller reservoirs behind small dams or between embankments: the **impounding** type collects and stores runoff; the **dug-out** type stores most of its water below ground level.
- The words tank and pond are often used interchangeably.

Development of tank irrigation

- Tank irrigation is an old, established practice in the semi-arid tropics, where erratic monsoon rain falls in only a few months; tanks store water and regulate it for farm and domestic use.
- In southern semi-arid India tanks date from Vedic times. The Ramayana mentions two: the Lake of Five Nymphs (Panchapraratataka), associated with Madakarni/Satkarni, and Pampasaras (Pampasagar) in Huvimothadagalli taluk, Bellary district, Karnataka, on the Tungabhadra.
- Many tanks in Tamil Nadu, Karnataka, Andhra Pradesh, Maharashtra and Madhya Pradesh have served for generations.
- Telangana was known as the "land of rice and tanks": Warangal and Karimnagar districts hold historic tanks such as Pakhal, Ramappa, Laknavaram and Sanigaram, built in the 12th–13th centuries.
- **System tanks:** a chain of tanks at different levels of one watershed, as at Katagiri (inscription dated 1096 A.D.); still popular in Salem district, Tamil Nadu.
- In 1853 R. Baird Smith counted about 43,000 tanks in repair and 10,000 out of repair in the Madras Presidency (≈ 53,000 in all).
- Tank irrigation received little attention in the Five-Year Plans (from 1951); in most districts of the former Madras Presidency its extent has hardly changed for over a century.

Lift irrigation

- Lift irrigation raises water from rivers, streams, canals, reservoirs and tanks. It is essential where the topography rules out gravity flow, yet India's lift potential remains under-exploited.
- It is as old as civilisation: the Hanging Gardens of Babylon (Iraq) were watered from the Euphrates with manually operated counterpoise lifts.
- Lifting **surface water** is cheaper than lifting groundwater — there is no well to build, and large volumes can be pumped.

1. Install pumps at or near the intake.
2. Deliver the water through a common delivery pipe, the **rising main**, in one stage or several depending on the site.

3. Distribute it in the command through open channels or underground pipelines.

Type	Where it fits
River lift irrigation	Where stream flow is adequate at peak irrigation demand
Canal lift irrigation	Land above the canal command; uses low- or medium-head, high-discharge pumps delivering to one or more distribution chambers and networks

- A lift scheme rarely pays for a single small farmer, so it is designed for groups of farmers and owned by government agencies or by cooperatives and group-farming societies, which get technical guidance from government and credit from financial institutions.
- Lift irrigation is widely used for rural water supply. Maharashtra has progressed rapidly through farmer cooperatives; Andhra Pradesh, Odisha, West Bengal and Uttar Pradesh are adopting it increasingly.

Sources: ICAR eCourse 'Minor Irrigation and Command Area Development' — Dr K. Shanmuga Sundaram (TNAU), reviewed by Dr P. Balakrishnan (UAS Karnataka). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Tanks built exclusively for groundwater recharge are called

(a) system tanks (b) percolation tanks
(c) dug-out ponds (d) balancing tanks

Q2. The region known as the "land of rice and tanks" is

(a) Konkan (b) Telangana
(c) Bundelkhand (d) Malwa

Q3. A chain of tanks at different levels of one watershed is called

(a) percolation tanks (b) system tanks
(c) anicuts (d) check dams

Q4. The common delivery pipe of a lift irrigation scheme is the

(a) suction pipe (b) rising main
(c) header (d) lateral

Q5. Lift irrigation from surface water is cheaper than from groundwater mainly because

(a) pumps are smaller (b) no well structure is needed
(c) water is cleaner (d) it needs no power

Answers

Q1 — (b) percolation tanks
Q2 — (b) Telangana
Q3 — (b) system tanks
Q4 — (b) rising main
Q5 — (b) no well structure is needed